iii) ***Missing data, if any may suitably be assumed.***

Marks

1. a) What is the difference between preemptive and non-preemptive scheduling algorithms? 2

b) Give the relative advantages and disadvantages of load time dynamic linking and run time dynamic linking. 2

c) Mention the two distinct parts of a filesystem. 2

d) If encryption can prove the identity of the sender of a message, then why do we need separate authentication algorithms? Mention the reasons. 2

e) What is web-site defacement? Under which breach category does it fall? 2

f) What are the kernel modules in linux? 2

2. a) What is a deadlock? What are the four conditions necessary for a deadlock to occur in a system? Explain each condition with a real world example. 7

b) Explain readers & writers problem. 5

3. a) What is the purpose of a TLB? Explain the TLB lookup with the help of a block diagram and the hardware required. 7

b) What is copy-on write feature, and under what circumstances is it beneficial to use this feature? What hardware support is required to implement this feature? 5

4. a) Give an example of an application that could benefit from operating system support for random access to indexed files. 6

	b)	Describe the implementation of access matrix using: Access control list, capability list compare both methods.	6	3
5.	a)	Explain how protection domains, combined with an access matrix are used to specify the resources a process may access.	6	2
	b)	How do viruses work?	6	1
6.	a)	Discuss the file system supported by windows 7 (NTFS, FAT32)	6	2
	b)	Explain the design principles of the Linux operating system.	6	2
7.	a)	Explain critical section problem in detail. What are the requirements for a solution to the critical section problem?	6	2
	b)	Consider the following page reference string: 1,2,3,4,2,1,5,6,2,1,2,3,7,6,3,2,1,2,3,6. How many page faults would occur for the following replacement algorithms assuming three and four frames? Initially all frames are empty. a) LRU replacement b) Optimal replacement.	6	3

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